Normalized Tables

Storage_Areas

StorageAreaID	Length	Width	Height	Location	JobsiteID	Total Stored	Is_Container

Materials

MaterialID	Name	Material Type	Length	Width	Height	Supplier Name	Total Available	Lost Amounts

Stored_In

StorageAreaID	MaterialID	Amount

Jobsites

JobsiteID	JobsiteName	JobsiteSupervisorID

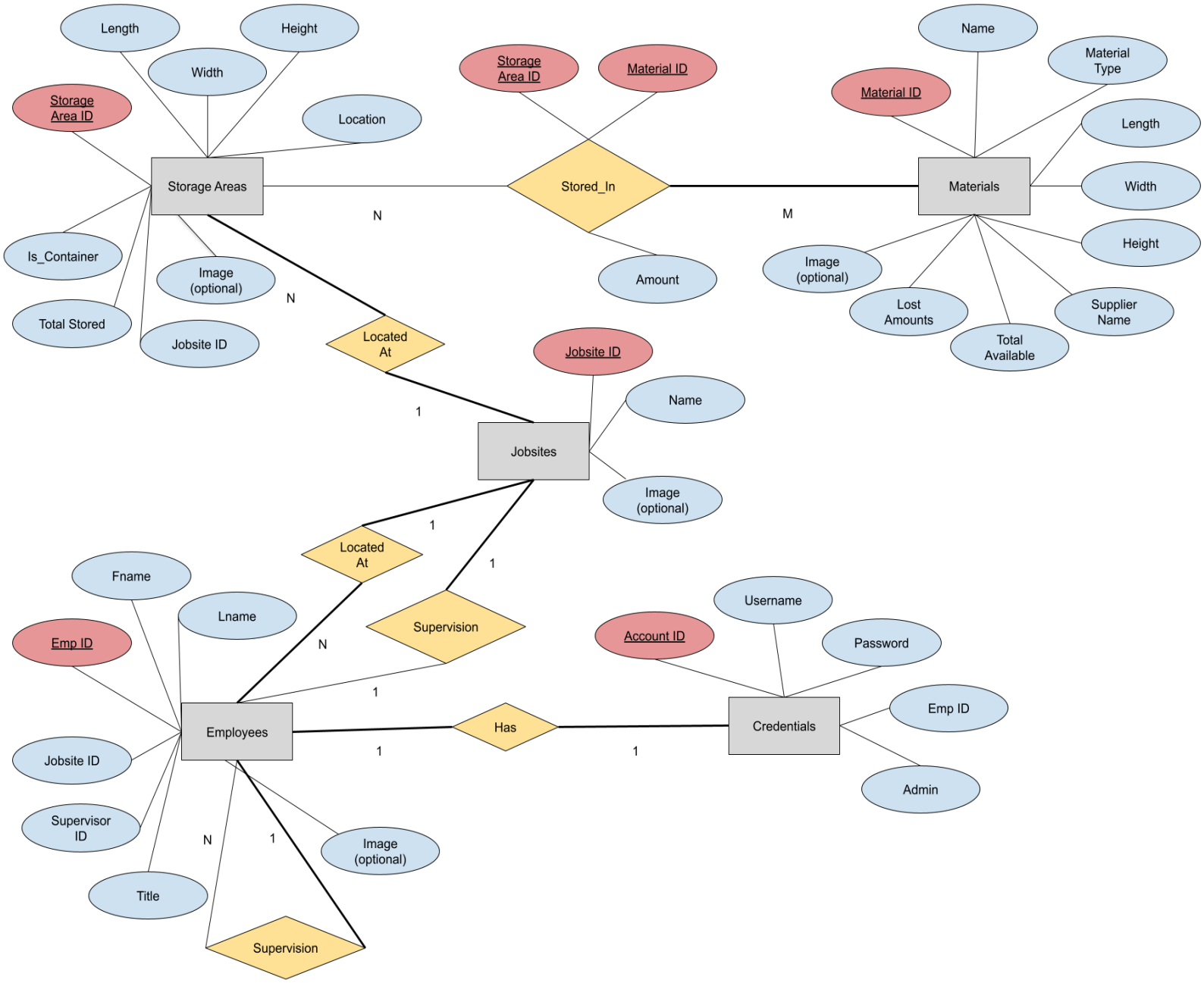
Employees

EmpID	Fname	Lname	Title	SupervisorID	JobsiteID

Credentials

AccountID	Username	Password	EmpID	Admin

Entity Relationship Diagram



Creation

CREATE TABLE Jobsite (JobsiteID INT PRIMARY KEY AUTO_INCREMENT, JobsiteName VARCHAR(32));

CREATE TABLE Employees (EmpID INT PRIMARY KEY AUTO_INCREMENT, Fname VARCHAR(32), Lname VARCHAR(32), Title VARCHAR(32), SupervisorID INT, JobsiteID INT, FOREIGN KEY(JobsiteID) REFERENCES Jobsite(JobsiteID) ON DELETE SET NULL) ;

ALTER TABLE Employees ADD FOREIGN KEY(SupervisorID) REFERENCES Employees(EmpID) ON DELETE SET NULL;

CREATE TABLE Material (MaterialID INT PRIMARY KEY AUTO_INCREMENT, Name VARCHAR(64), MaterialType VARCHAR(64), Length DECIMAL(5,2), Width DECIMAL(5,2), Height DECIMAL(5,2), SupplierName VARCHAR(64), TotalAvailable SMALLINT, LostAmounts SMALLINT);

CREATE TABLE Storage_Area (StorageAreaID INT PRIMARY KEY AUTO_INCREMENT, Length DECIMAL(5,2), Width DECIMAL(5,2), Height DECIMAL(5,2), Location VARCHAR(64), JobsiteID INT, TotalStored SMALLINT, Is_Container BOOLEAN, FOREIGN KEY(JobsiteID) REFERENCES Jobsite(JobsiteID) ON DELETE SET NULL);

CREATE TABLE Stored_In (StorageAreaID INT, MaterialID INT, Amount SMALLINT, PRIMARY KEY (StorageAreaID, MaterialID));

CREATE TABLE Credentials (AccountID INT PRIMARY KEY AUTO_INCREMENT, Username VARCHAR(32), Password VARCHAR(32), EmpID INT, Admin BOOLEAN DEFAULT FALSE, FOREIGN KEY(EmpID) REFERENCES Employees(EmpID) ON DELETE CASCADE);

For optional images add "**, Image VARCHAR(256)**" where desired. This should be a link to a repository where the image is saved.

Sample Entries For Database

Employees

EmpID	Fname	Lname	Title	SupervisorID	JobsiteID
1	George	Washington	Owner	1	NULL
2	John	Adams	Site Supervisor	1	1
3	Thomas	Jefferson	Site Supervisor	1	2
4	James	Madison	Foreman	2	1
5	James	Monroe	Insulator	4	1
6	John Q.	Adams	Insulator	4	1
7	Andrew	Jackson	Foreman	3	2
8	Martin	Van Buren	Insulator	7	2
9	William H.	Harrison	Insulator	7	2
10	John	Tyler	Insulator	7	2
11	James K.	Polk	Helper	4	1
12	Zachary	Taylor	Helper	7	2
13	Millard	Fillmore	Insulator	4	1

INSERT INTO Employees(Fname, LName, Title, SupervisorID, JobsiteID) VALUES("George", "Washington", "Owner", 1, NULL);
INSERT INTO Employees(Fname, LName, Title, SupervisorID, JobsiteID) VALUES("John", "Adams", "Site Supervisor", 1, 1);
INSERT INTO Employees(Fname, LName, Title, SupervisorID, JobsiteID) VALUES("Thomas", "Jefferson", "Site Supervisor", 1, 2);
INSERT INTO Employees(Fname, LName, Title, SupervisorID, JobsiteID) VALUES("James", "Madison", "Foreman", 2, 1);
INSERT INTO Employees(Fname, LName, Title, SupervisorID, JobsiteID) VALUES("James", "Monroe", "Insulator", 4, 1);
INSERT INTO Employees(Fname, LName, Title, SupervisorID, JobsiteID) VALUES("John Q.", "Adams", "Insulator", 4, 1);

INSERT INTO Employees(Fname, LName, Title, SupervisorID, JobsiteID) VALUES("Andrew", "Jackson", "Foreman", 3, 2);
INSERT INTO Employees(Fname, LName, Title, SupervisorID, JobsiteID) VALUES("Martin", "Van Buren", "Insulator", 7, 2);
INSERT INTO Employees(Fname, LName, Title, SupervisorID, JobsiteID) VALUES("William H.", "Harrison", "Insulator", 7, 2);
INSERT INTO Employees(Fname, LName, Title, SupervisorID, JobsiteID) VALUES("John", "Tyler", "Insulator", 7, 2);
INSERT INTO Employees(Fname, LName, Title, SupervisorID, JobsiteID) VALUES("James K.", "Polk", "Helper", 4, 1);
INSERT INTO Employees(Fname, LName, Title, SupervisorID, JobsiteID) VALUES("Zachary", "Taylor", "Helper", 7, 2);
INSERT INTO Employees(Fname, LName, Title, SupervisorID, JobsiteID) VALUES("Millard", "Fillmore", "Insulator", 4, 1);

Storage Areas*

StorageAreaID	Length	Width	Height	Location	JobsiteID	TotalStored	Is_Container
1	30.48	30.48	7.62	Laydown 1	1	10	False
2	15.24	15.24	7.62	Laydown 2	2	20	False
3	12.2	2.4	2.6	Laydown 1	1	20	True
4	12.2	2.4	2.6	Laydown 1	1	25	True

INSERT INTO Storage_Area(Length, Width, Height, Location, JobsiteID, TotalStored, Is_Container) VALUES (30.48, 30.48, 7.62, “Laydown 1”, 1, 10, False);
INSERT INTO Storage_Area(Length, Width, Height, Location, JobsiteID, TotalStored, Is_Container) VALUES (15.24, 15.24, 7.62, "Laydown 2", 2, 20, False);
INSERT INTO Storage_Area(Length, Width, Height, Location, JobsiteID, TotalStored, Is_Container) VALUES (12.2, 2.4, 2.6, "Laydown 1", 1, 20, True);
INSERT INTO Storage_Area(Length, Width, Height, Location, JobsiteID, TotalStored, Is_Container) VALUES (12.2, 2.4, 2.6, "Laydown 1", 1, 25, True);

Materials*

Material ID	Name	Material Type	Length	Width	Height	Supplier Name	TotalAvailable	LostAmounts
1	"Insulation 1"	"Insulation"	0.92	0.92	0.92	"Brand 1"	10	0
2	"Insulation 2"	"Insulation"	0.92	0.92	0.61	"Brand 1"	30	0
3	"Insulation 3"	"Insulation"	1.22	1.22	1.22	"Brand 2"	25	10
4	"Tape 1"	"Tape"	0.3	0.3	0.3	"Brand 3"	10	0

INSERT INTO Material(Name, MaterialType, Length, Width, Height, SupplierName, TotalAvailable, LostAmounts) VALUES("Insulation 1", "Insulation", 0.92, 0.92, 0.92, "Brand 1", 10, 0);
INSERT INTO Material(Name, MaterialType, Length, Width, Height, SupplierName, TotalAvailable, LostAmounts) VALUES("Insulation 2", "Insulation", 0.92, 0.92, 0.61, "Brand 1", 30, 0);
INSERT INTO Material(Name, MaterialType, Length, Width, Height, SupplierName, TotalAvailable, LostAmounts) VALUES("Insulation 3", "Insulation", 1.22, 1.22, 1.22, "Brand 2", 25, 10);
INSERT INTO Material(Name, MaterialType, Length, Width, Height, SupplierName, TotalAvailable, LostAmounts) VALUES("Tape 1", "Tape", 0.3, 0.3, 0.3, "Brand 3", 10, 0);

Jobsites

JobsiteID	JobsiteName
1	ABC Data Center
2	XYZ Refinery

INSERT INTO Jobsite(JobsiteName) VALUES(“ABC Data Center”);
INSERT INTO Jobsite(JobsiteName) VALUES("XYZ Refinery");

Stored_In

StorageAreaID	MaterialID	Amount
1	4	10
2	2	20
3	1	10
3	3	10
4	3	15
4	2	10

```
INSERT INTO Stored_In(StorageAreaID, MaterialID, Amount) Values(1, 4, 10);
INSERT INTO Stored_In(StorageAreaID, MaterialID, Amount) Values(2, 2, 20);
INSERT INTO Stored_In(StorageAreaID, MaterialID, Amount) Values(3, 1, 10);
INSERT INTO Stored_In(StorageAreaID, MaterialID, Amount) Values(3, 3, 10);
INSERT INTO Stored_In(StorageAreaID, MaterialID, Amount) Values(4, 3, 15);
INSERT INTO Stored_In(StorageAreaID, MaterialID, Amount) Values(4, 2, 10);
```

Credentials

AccountID	Username	Password	EmpID	Admin
1	GWashington	April1789	1	TRUE
2	JAdams	March1797	2	TRUE
3	TJefferson	March1801	3	TRUE
4	JMadison	March1809	4	FALSE
5	JMonroe	March1817	5	FALSE
6	JQAdams	March1825	6	FALSE
7	AJackson	March1829	7	FALSE
8	MVanBuren	March1837	8	FALSE
9	WHHarrison	March1841	9	FALSE
10	JTyler	April1841	10	FALSE
11	JKPolk	March1845	11	FALSE

12	ZTaylor	March1849	12	FALSE
13	MFillmore	July1850	13	FALSE

INSERT INTO Credentials(Username, Password, EmpID, Admin) VALUES("GWashington", "April1789", 1, TRUE);
INSERT INTO Credentials(Username, Password, EmpID, Admin) VALUES("JAdams", "March1797", 2, TRUE);
INSERT INTO Credentials(Username, Password, EmpID, Admin) VALUES(“TJefferson”, “March1801”, 3, TRUE);
INSERT INTO Credentials(Username, Password, EmpID) VALUES("JMadison", “March1809”, 4);
INSERT INTO Credentials(Username, Password, EmpID) VALUES("JMonroe", "March1817", 5);
INSERT INTO Credentials(Username, Password, EmpID) VALUES("JQAdams", “March1825”, 6);
INSERT INTO Credentials(Username, Password, EmpID) VALUES("AJackson", "March1829", 7);
INSERT INTO Credentials(Username, Password, EmpID) VALUES("MVanBuren", "March1837", 8);
INSERT INTO Credentials(Username, Password, EmpID) VALUES("WHHarrison", "March1841", 9);
INSERT INTO Credentials(Username, Password, EmpID) VALUES("JTyler", "March1845", 10);
INSERT INTO Credentials(Username, Password, EmpID) VALUES("JKPolk", “March1845”, 11);
INSERT INTO Credentials(Username, Password, EmpID) VALUES("ZTaylor", "March1849", 12);
INSERT INTO Credentials(Username, Password, EmpID) VALUES("MFillmore", "July1850", 13);

*Length, width, and height are all in meters in these examples. Conversions to other measurements should be handled on either the frontend or backend.

Useful Queries

Get everything from table:
SELECT * FROM Storage_Areas;

Get Leadership:
`SELECT DISTINCT emp1.empid AS EmpID, CONCAT(emp1.fname, ' ', emp1.lname) AS Name, emp1.title AS Title
FROM employees as emp1
JOIN employees as emp2
ON emp2.supervisorid = emp1.empid
LIMIT <num> OFFSET <num>;`

Get subordinates:
SELECT * FROM employees WHERE SupervisorID = <num>;

Get employees at jobsites:
`SELECT employees.empid AS EmpID, employees.fname AS FirstName, employees.lname AS LastName, employees.title AS Title,
jobsites.jobsitename AS Jobsite
FROM employees
LEFT JOIN jobsites
ON employees.jobsiteid = jobsites.jobsiteid
LIMIT <num> OFFSET <num>;`

Get material amounts:
`SELECT materials.name AS Name, materials.suppliername AS Brand,
storage_areas.location AS Location,
jobsites.jobsitename AS Jobsite,
stored_in.amount AS Amount
FROM materials
JOIN stored_in
ON materials.materialid = stored_in.materialid
JOIN storage_areas
ON storage_areas.storageareaid = stored_in.storageareaid
JOIN jobsites
ON jobsites.jobsiteid = storage_areas.jobsiteid
ORDER BY materials.name
LIMIT <num> OFFSET <num>;`

Get the materials with the lowest available amounts:
SELECT * FROM materials ORDER BY totalavailable ASC LIMIT <num>;